



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Name : Atharva Ambilwade

PRN : 25070122055

Class : CSE

Batch : 2025-29

Subject: Programming with Java

ASSIGNMENT No : 11

TITLE : Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.



Code and Output Screenshots

```
J M1.java > Java > M1
1 interface Animal {
2     void sound();
3 }
4
5 class Horse implements Animal {
6     public void sound() {
7         System.out.println("Horse neighs.");
8     }
9 }
10
11 public class M1 {
12     Run main | Debug main | Run | Debug
13     public static void main (String[] args) {
14         Animal a = new Horse();
15         a.sound();
16     }
17 }
```

PROBLEMS 22 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...) Code

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac M1.java && java M1
Horse neighs.

[Done] exited with code=0 in 0.494 seconds
```

Conclusion

This program demonstrates the use of a Java interface. The Horse class implements the Animal interface by providing the implementation of the sound() method. This shows how interfaces define a common behavior while allowing different classes to provide their own implementations.



Exercise

1. Create an interface Printable and implement it in Student and Employee classes.

```
J M2.java > Java > M2
1  interface Printable {
2      void print();
3  }
4  class Student implements Printable {
5      public void print() {
6          System.out.println("Student Details");
7      }
8  }
9  class Employee implements Printable {
10     public void print() {
11         System.out.println("Employee Details");
12     }
13 }
14 public class M2 {
    Run main | Debug main | Run | Debug
15     public static void main(String[] args) {
16         Student s = new Student();
17         Employee e = new Employee();
18         System.out.println("Student object : ");
19         s.print();
20         System.out.println("Employee object : ");
21         e.print();
22     }
23 }
24 }
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, !excludeText, t...

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac M2.java && java M2
Student object :
Student Details
Employee object :
Employee Details
|
[Done] exited with code=0 in 0.486 seconds
```



2. Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

```
J M3.java > Java > M3
1  interface Switchable {
2      void turnOn();
3  }
4  class Light implements Switchable {
5      public void turnOn() {
6          System.out.println("Light is now ON");
7      }
8  }
9  class Fan implements Switchable {
10     public void turnOn() {
11         System.out.println("Fan is now ON");
12     }
13 }
14 public class M3 {
15     Run main | Debug main | Run | Debug
16     public static void main(String[] args) {
17         Light l = new Light();
18         Fan f = new Fan();
19         l.turnOn();
20         f.turnOn();
21     }
22 }
```

PROBLEMS 26

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, excludeText, t...

[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac M3.java && java M3

Light is now ON

Fan is now ON

[Done] exited with code=0 in 0.478 seconds